

Low-Temperature Long Fermentation Effects on Ginjo Aroma Formation in Junmai Sake

Tanaka et al. (2024, fictional)

1. Introduction

Low-temperature, long-duration fermentation is known to strongly influence the aroma compounds of ginjo-grade sake. This study examines how fermentation temperature affects isoamyl acetate concentration in junmai moromi. Isoamyl acetate, the principal component of ginjo aroma, is produced along a yeast metabolic pathway, and lower fermentation temperatures have been reported to increase its yield.

2. Methods

Two fermentation conditions were compared: 8 degrees C and 15 degrees C. The 8-degree batch fermented for 25 days and the 15-degree batch for 18 days, with the moromi sampled daily. Yeast strain Kyokai No. 901 was used, with Gohyakumangoku rice milled to a 60% seimaibuai.

3. Results and discussion

The batch fermented at 8 degrees C for 25 days reached isoamyl acetate concentrations approximately 1.8 times higher than the batch fermented at 15 degrees C for 18 days. However, the low-temperature long-fermentation batch showed slower progression of sake meter value and a longer time to pressing, yielding a somewhat lower recovery. Low-temperature fermentation raises ginjo aroma output but must be weighed against the yield trade-off.